

Endometriosis Pharmacotherapy

Based on Pharmacotherapy Handbook, 7th Edition
Prepared for Pharmacy Yr IV UG Students

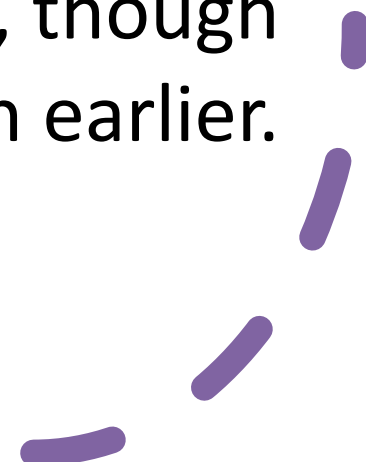
Introduction

- Endometriosis is a chronic condition where tissue similar to the lining of the uterus (endometrium) grows outside the uterus, most often in the pelvis, causing pain, heavy periods, and sometimes infertility.



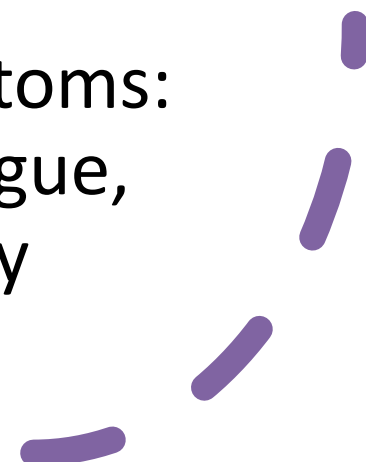


Epidemiology

- Affects about **10% of reproductive-age women worldwide** ($\approx$ 190 million women).
 - Often diagnosed in women in their 30s and 40s, though symptoms can begin earlier.
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Symptoms

- Severe menstrual pain (dysmenorrhea)
 - Heavy menstrual bleeding
 - Chronic pelvic pain (not limited to menstruation)
 - Pain during sexual intercourse, bowel movements, or urination
 - Infertility
 - Other systemic symptoms: bloating, nausea, fatigue, depression, or anxiety
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Diagnosis of Endometriosis

1. Clinical Diagnosis (Primary Method)

- Endometriosis is **primarily a clinical diagnosis** based on:
- **Typical Symptoms**
- **Chronic pelvic pain** (>6 months)
- **Dysmenorrhea** (progressive and severe)
- **Dyspareunia** (especially deep)
- **Infertility**
- **Dyschezia**, painful bowel movements
- **Cyclic urinary symptoms** (if bladder involvement)
- **Pain out of proportion** to exam findings is a common clue.

Diagnosis of Endometriosis

2. Physical Examination Findings

- Most women have a **normal exam**, but possible findings include:
- Uterosacral ligament nodularity
- Fixed, retroverted uterus
- Adnexal mass (endometrioma)
- Tenderness in the cul-de-sac
- Decreased uterine mobility
- Exam is more useful in **moderate–severe disease**.

Diagnosis of Endometriosis

3. Imaging (Supports Diagnosis, Does NOT Confirm)

— a) Transvaginal Ultrasound (TVUS) — FIRST LINE

- Best for detecting:
- **Endometriomas** (classic “ground-glass” cyst)
- Deep infiltrating lesions in the rectovaginal space (with expert sonographers)
- **Note:** TVUS cannot reliably detect superficial peritoneal disease.

— b) MRI

- Used when:
- Deep infiltrating endometriosis (DIE) is suspected
- Bowel, bladder, or ureter involvement
- Ultrasound is inconclusive
- MRI gives excellent mapping prior to surgery.

Diagnosis of Endometriosis

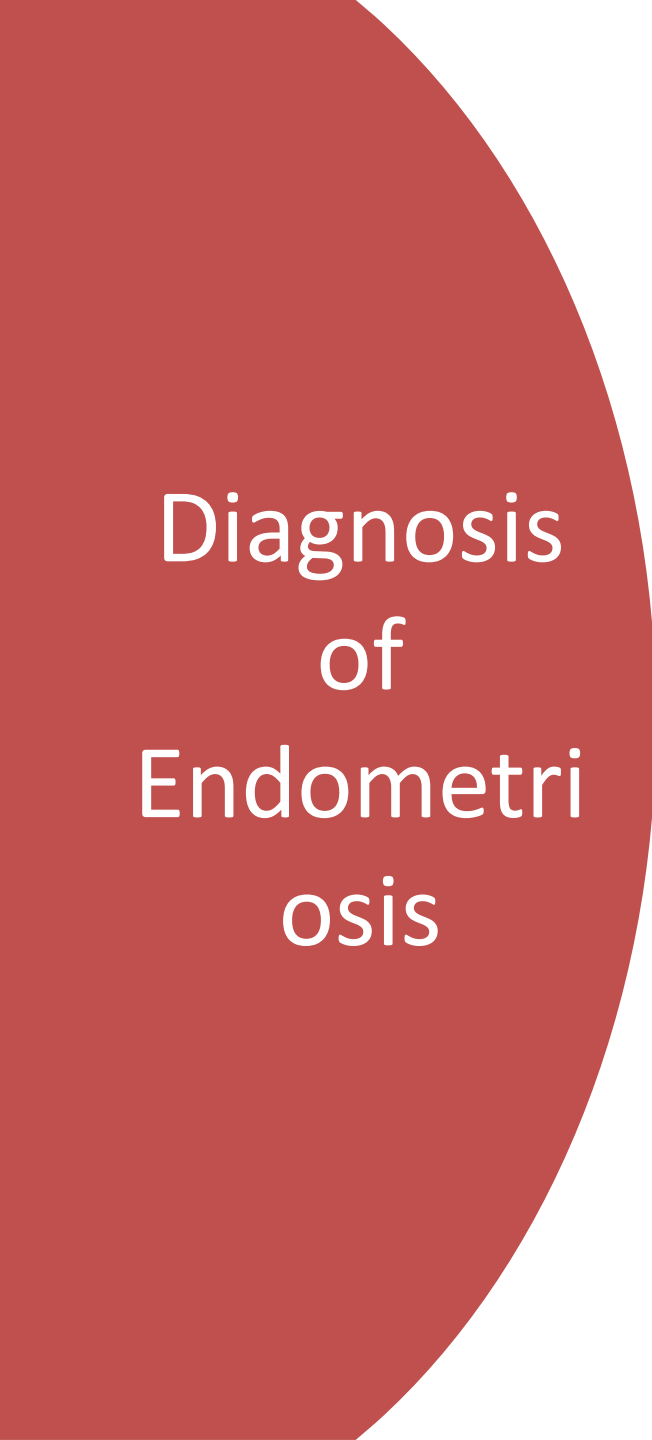
4. Laboratory Tests

- **CA-125 (Cancer Antigen 125)** may be elevated but is **not diagnostic**.
- Normal CA-125 **does NOT rule out** endometriosis.
- Useful only in:
 - Monitoring large endometriomas
 - Severe disease
 - Differentiating from other pelvic masses

Diagnosis of Endometriosis

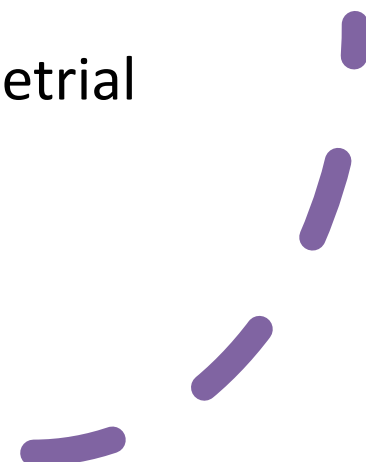
5. Response to Empiric Treatment

- ACOG and ESHRE support **empirical treatment** for suspected endometriosis:
 - NSAIDs
 - Combined oral contraceptives
 - Progestin therapy
 - LNG-IUS
 - GnRH agonists/antagonists
 - Improvement with treatment supports—but does not confirm—the diagnosis.

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Diagnosis of Endometriosis

6. Definitive Diagnosis (Gold Standard):

- **Laparoscopy with Histologic Confirmation**
 - The only **definitive** diagnosis requires:
 - Visualization of lesions at laparoscopy
 - **Biopsy** showing endometrial glands and/or stroma
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Goals of Therapy

- - Reduce pain and inflammation
- - Suppress ectopic endometrial tissue growth
- - Improve quality of life
- - Preserve fertility when possible

Treatment



- **No cure currently exists.**
- Management focuses on symptom control and improving quality of life:
 - Pain relief (NSAIDs, hormonal therapy such as oral contraceptives, progestins, or GnRH agonists).
 - Surgery (laparoscopic excision or ablation of lesions).
 - Fertility support (assisted reproductive technologies if needed).
- Multidisciplinary care is often required, including gynecology, pain management, and mental health support.

First-Line Medical Therapies



These are preferred due to effectiveness, safety, and cost.

a) NSAIDs

- Used for **pain relief**, not hormone suppression
- Ibuprofen, naproxen, diclofenac
- Best for dysmenorrhea
- Often combined with hormonal therapies

b) Combined Hormonal Contraceptives (CHCs)

(Combined oral pills, patch, ring)

- Reduce menstrual flow → lower retrograde menstruation
- Decrease prostaglandin-mediated pain
- Continuous use is more effective than cyclic
- ACOG & ESHRE first-line for pain

2. Progestin-Only Therapies

☛ These suppress ovulation and thin endometrium.

a) Oral Progestins

- **Dienogest (2 mg daily)** → ESHRE prefers this
- **Norethindrone acetate**
- **Medroxyprogesterone acetate (oral)**

b) Long-Acting Progestins

- **Depot medroxyprogesterone acetate (DMPA)**
- Effective for pain, but may ↓ bone mineral density (long-term use caution)

c) Levonorgestrel-Releasing IUS (LNG-IUS)

- Excellent for dysmenorrhea and chronic pelvic pain
- Reduces lesion activity
- ACOG & ESHRE strongly recommend for pain control
- Common first-line option for those not seeking fertility

3. Second-Line Therapies

- Used when first-line treatments fail or are not tolerated.

a) GnRH Agonists (e.g., Leuprolide, Goserelin, Nafarelin)

- Induce **hypoestrogenic state** (medical menopause)
- Effective for moderate to severe pain
- Cause hot flashes, bone loss
- Must use **add-back therapy** (progestin or low-dose estrogen-progestin) to prevent BMD loss

b) GnRH Antagonists (e.g., Elagolix, Relugolix)

- Oral, fast-acting suppression
- Dose-dependent estrogen reduction
- Fewer menopausal symptoms than agonists
- Also require monitoring for BMD
- Preferred over agonists in some guidelines for ease of use

4. Third-Line / Adjunctive Therapies

a) Danazol (Androgenic)

- Effective but rarely used now
- Adverse effects: acne, weight gain, adverse lipid changes, hirsutism
- Contraindicated in hyperlipidemia, liver disease

b) Aromatase Inhibitors (Letrozole, Anastrozole)

- Reduce local estrogen production within lesions
- For **severe refractory disease**
- Used **only in combination** with progestins or GnRH analogs (to prevent ovarian stimulation)

Clinical Notes (ACOG & ESHRE)

(Dr. Tamrat)

- **Start with medical therapy** unless contraindicated.
- **Surgery** is reserved for refractory pain or infertility.
- **LNG-IUS, dienogest, and CHCs** are top choices for long-term management.
- Long-term GnRH agonists/antagonists require **bone protection**.

References

- Pharmacotherapy Handbook, 7th Edition – Chapter 31.
 - ACOG Practice Bulletin: Management of Menopausal Symptoms, 2023.
 - North American Menopause Society (NAMS) Guidelines, 2023.
 - ACOG Bulletin on Abnormal Uterine Bleeding, 2024.
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